

Convection Heat Transfer Tutorial

1 Questions

1. Air at 27 °C and 1 atm flows over a flat plate at a speed of 2 m/s. Calculate the boundary layer thickness at distances of 20 cm and 40 cm from the leading edge of the plate. Calculate the mass flow that enters the boundary layer between this distance. Assume unit depth in the z direction. [0.56 cm, 0.8 cm, $3.4 \times 10^{-3} \text{ kg/s}$].
2. Air at 27 °C and 1 atm flows over a flat plate at a speed of 2 m/s. The plate is maintained at 60 °C. Calculate the heat transferred in the first 20 cm, and 40 cm. [81 W, 115 W].
3. Engine oil at 20 °C flows over a 20 cm square plate at a velocity of 1.2 m/s. The plate is heated to a uniform temperature of 60 °C. Calculate the heat lost by the plate. [350.6 W]
4. Air at 20 °C and 1 atm flows over a flat plate at 35 m/s. The plate is 75 cm long and is maintained at 60 °C. Assuming unit depth in the z-direction calculate heat transfer from the plate. [2373 W]
5. Air at 1 atm, 20 °C, is heated as it flows through a tube with a diameter of 2.54 cm, at a velocity of 20 m/s. Calculate the heat transfer / unit length of the tube if a constant heat flux condition is maintained at the wall and the wall temperature is 20 °C above the air temperature all along the length of the tube. How much would be the air bulk temperature rise over a length of 3 m length of the tube ?.
6. Air at 2 atm 200 °C, is heated as it flows through a tube with a diameter of 2.54 cm at a velocity of 20 m/s. Calculate the heat transfer per unit length of tube if a constant heat flux condition is maintained at the wall and the wall temperature is 20 °C above the air temperature, all along the length of the tube. How much would the bulk temperature increase over a 3m length of the tube ?.
7. Water at 60 °C enters a tube of 2.54 cm diameter at a mean flow velocity of 2 cm/s. Calculate the exit water temperature if the tube is 3 m long and the wall temperature is constant at 80 °C.

8. Water flows with a mean velocity of 2 m/s inside a circular pipe of inside diameter 5 cm. The pipe is of commercial steel with relative roughness 0.0009 and its wall is maintained at a uniform temperature 100 °C by condensing steam on its outer surface. At a location where the fluid is hydro dynamically and thermally developed, the bulk mean temperature of water is 60 °C. Calculate the heat transfer coefficient by using the Petukhov equation. Also calculate the heat transfer coefficient assuming smooth pipe using (i) Petukhov equation (ii) The Sieder and Tate equation (iii) The Dittus Boelter equation.
9. A fine wire having a diameter of 3.94×10^{-5} m is placed in a 1 atm air stream at 25 °C having a flow velocity of 50 m/s perpendicular to the wire. An electric current is passed through the wire, raising the surface temperature to 50 °C. Calculate the heat loss per unit length.
10. Air at 1 atm and 10 °C flows across a bank of tubes 15 rows high and 5 row deep at a velocity of 7 m/s measured at a point in the flow before the air enters the tube bank. The surface of the tubes are maintained at 65 °C. The diameter of the tubes are 2.54 cm, they are arranged in an in-line manners so that the spacing in both the parallel directions to the flow is 3.81 cm. Calculate the total heat transfer per unit length for the tube bank and the exit air temperature.
11. A large vertical plate 4.0 m high is maintained at 60 °C and exposed to atmospheric air at 10 °C. Calculate the heat transfer if the plate is 10 m wide.
12. A 2.0 cm dia horizontal tube is maintained at a surface temperature of 38 °C by an internal electric heater and submerged in water at 27 °C. Calculate the free convection heat loss / unit length of the tube.